

An aerial photograph of the Dalhousie University campus in Halifax, Nova Scotia. The image shows a mix of modern and traditional architecture. In the foreground, a large, multi-story brick building with a central clock tower and a dome is prominent. To its right is a large green soccer field with white markings. In the background, there are several modern, multi-story buildings with glass facades. The campus is surrounded by trees and greenery.

CSCI 3171 - Network Computing

Packet vs Circuit Switching

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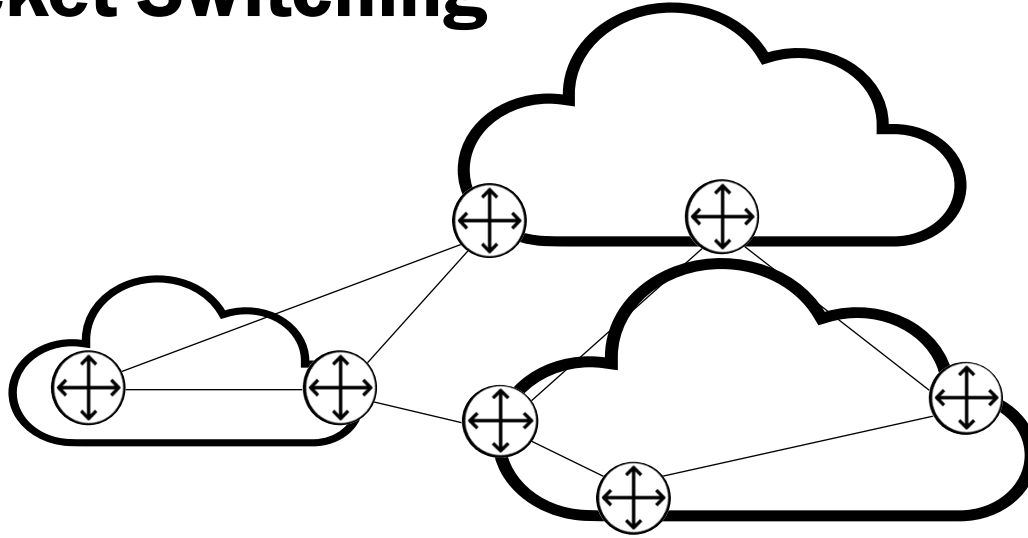


Goal and Roadmap

- Goal:
 - *Big picture* of the Internet
 - Introduction to terminology
- Overview and roadmap
 - What is the Internet? What is a protocol?
 - Network edge: hosts, access network, physical media
 - Network core: packet switching
 - Performance: loss, delay, throughput
 - Protocol layers, service models
 - *Packet and circuit switching*



Packet Switching

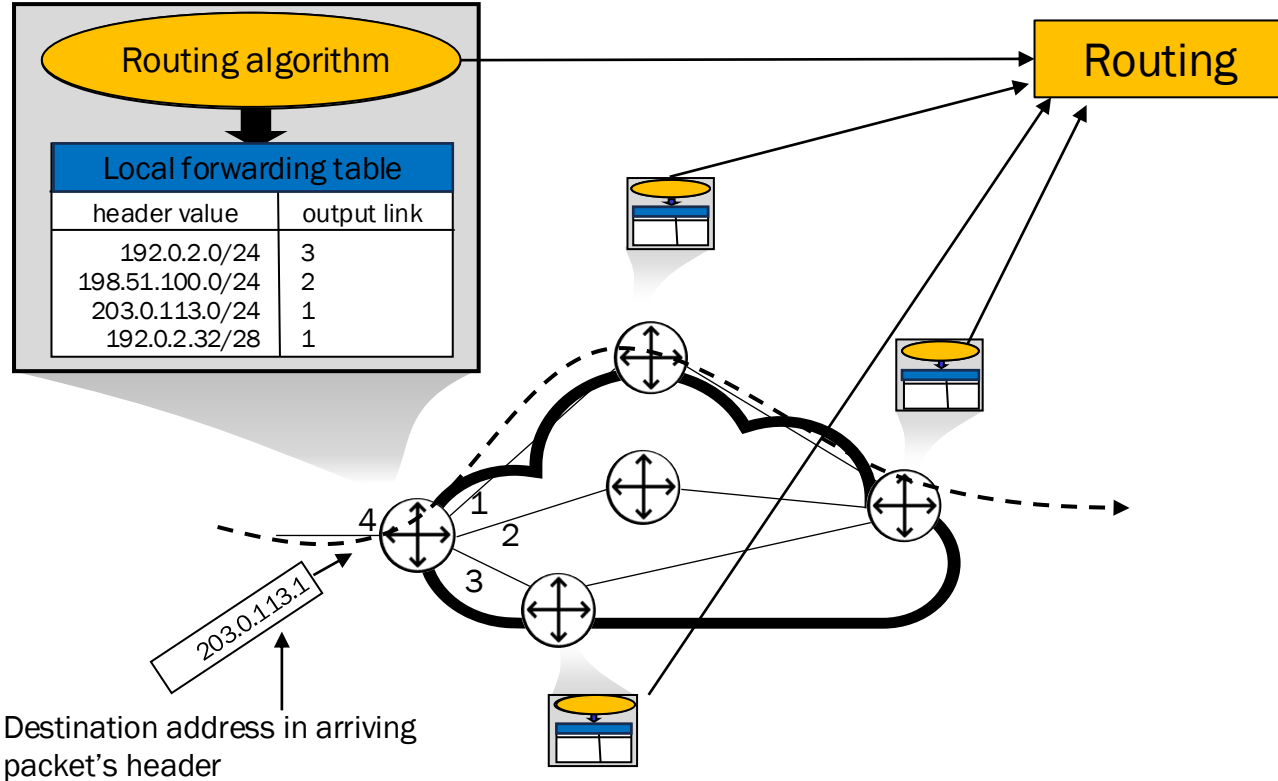


- Packet-switching: hosts break application-layer messages into packets
- Network forwards packets from one router to the next, across links on path from source to destination



Routing and Forwarding

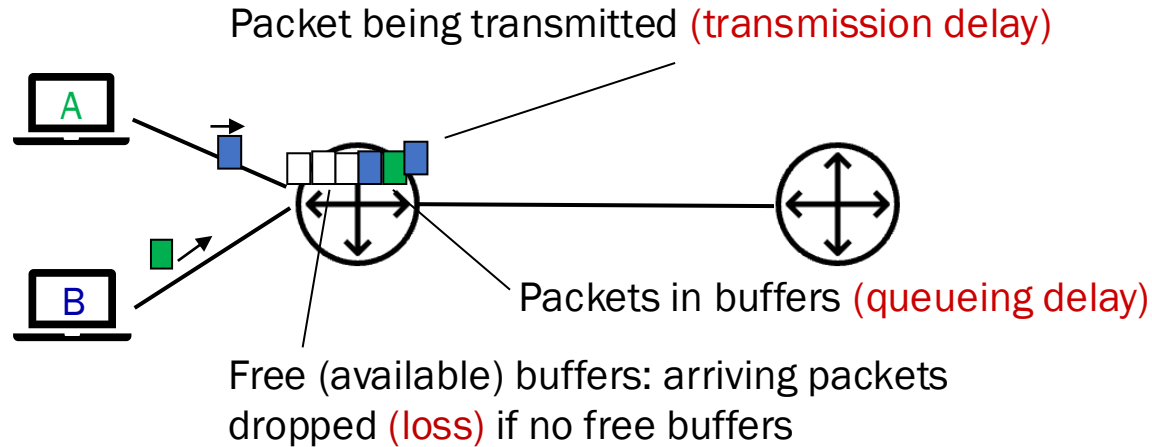
Forwarding





How Do Packet Delay and Loss Occur?

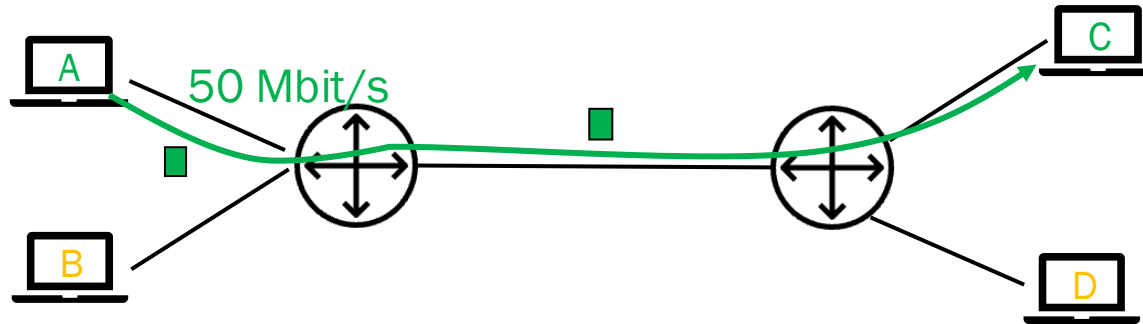
- Packets queue in router buffers, waiting for turn for transmission
 - Queue length grows when arrival rate to link (temporarily) exceeds output link transmission rate
 - Packet loss occurs when memory to hold queued packets fills up





Circuit Switching: an Alternative to Packet Switching

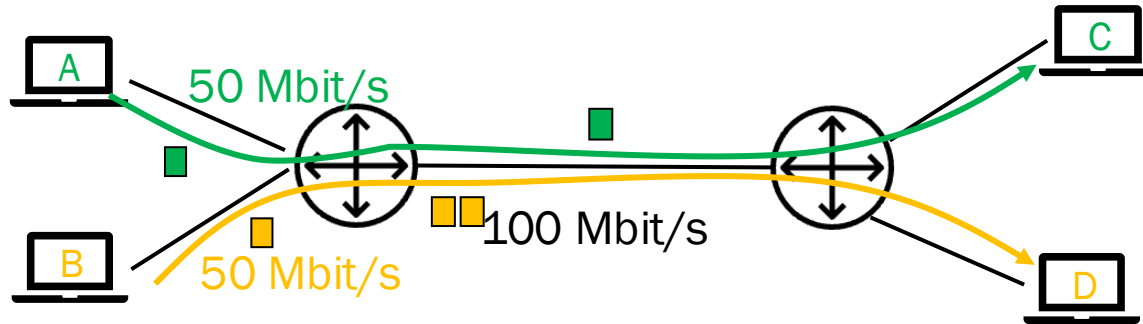
- Connection or *circuit* between source and destination
 - Connection establishment and reservation of resources
 - Dedicated resources for the circuit
 - Forwarding based on circuit identifier





Circuit Switching: The Good

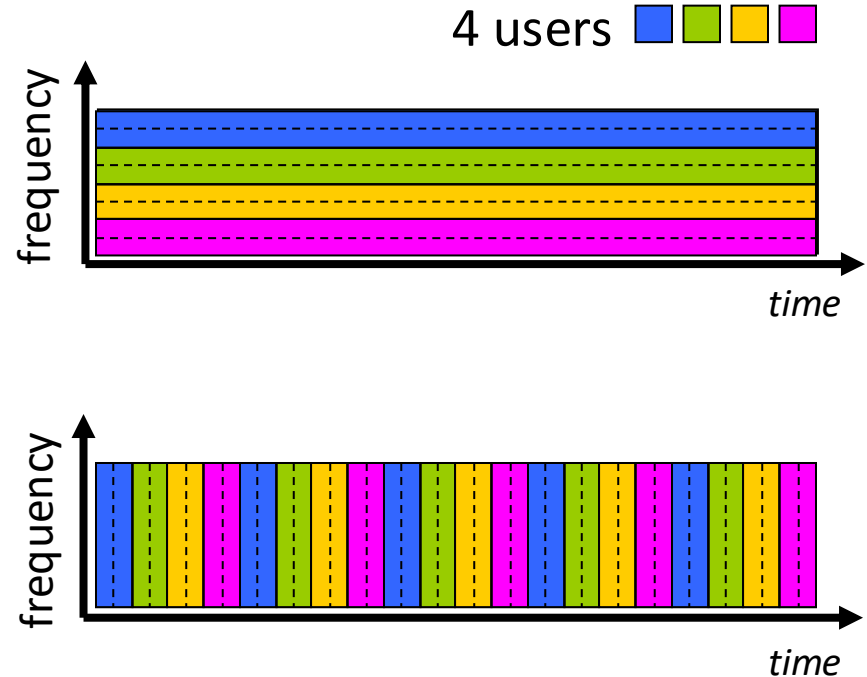
- Transmission rate for all links is equal to 100 Mbit/s
- Two *connections* requiring each 50 Mbit/s
 - Resources reserved for each connection
- Predictable performance: no delay, no packet loss





Circuit Switching: FDM and TDM

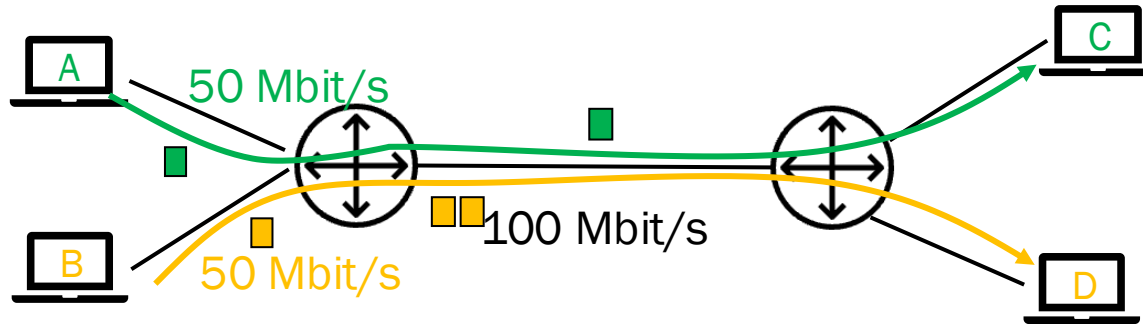
- Frequency Division Multiplexing (FDM)
 - Optical, electromagnetic frequencies divided into frequency bands
 - Each call allocated its own band
- Time Division Multiplexing (TDM)
 - Time divided into slots
 - Each call allocated periodic slots





Circuit Switching: The Bad

- Transmission rate for all links is equal to 100 Mbit/s
- Two *connections* requiring each 50 Mbit/s
- What happens when one connection is silent?





Circuit Switching

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Mentimeter

Two connections of 50 Mbit/s share a 100 Mbit/s bottleneck link. One connection stops sending traffic. The other one can use the total 100 Mbit/s.



■	True	✗
■	False	✓

→ Show correct answer

Menti

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Two connections of 50 Mbit/s share a 100 Mbit/s bottleneck link. One connection stops sending traffic. The other one can use the total 100 Mbit/s.

True ✗

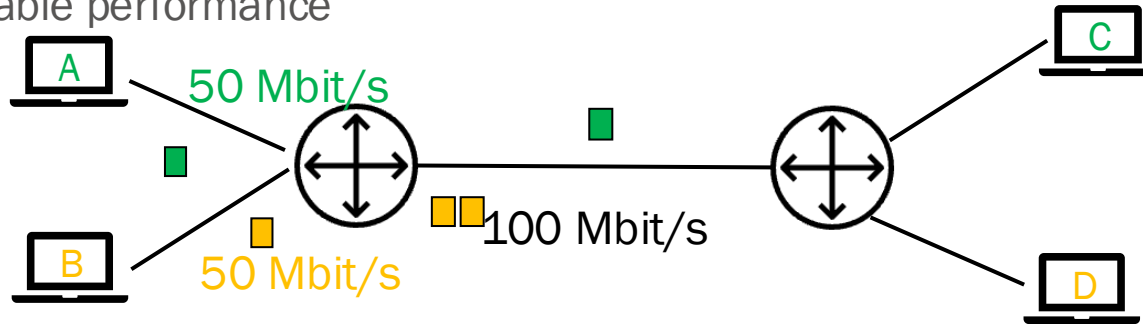
False ✓

Can you name an application you use over the Internet?



Comparison with Packet Switching

- What happens when one source is silent?
 - A is silent, B can send up to 100 Mbit/s
 - System efficiency
- What happens when it becomes active again?
 - B is sending 100 Mbit/s, A sends 50 Mbit/s => queuing delay and loss
 - Unpredictable performance





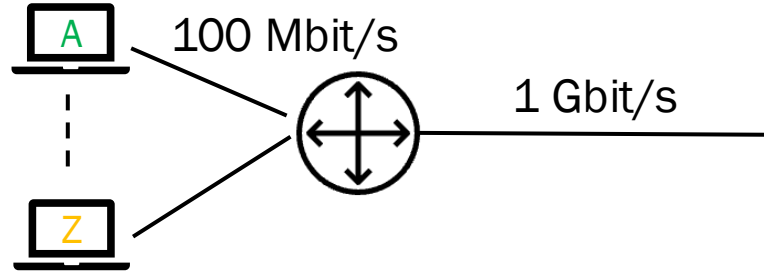
Circuit Switching: The Ugly

- Which one is easier to implement?
 - Packet switching or circuit switching
- Circuit switching necessitates:
 - Resource reservation, signaling, admission control
- Packet switching necessitates:
 - Congestion control



Packet Switching versus Circuit Switching

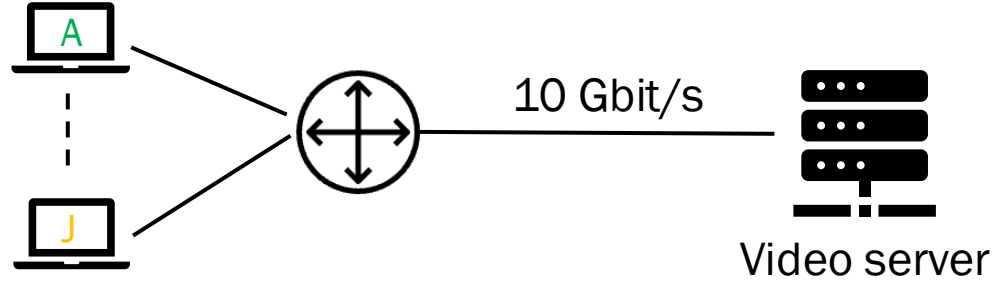
- 1 Gbit/s link
- Each user:
 - 100 Mbit/s when active
 - Active 10% of time
- How many users can use this network under circuit-switching and packet switching?
 - Circuit-switching: 10 users
 - Packet switching: with 35 users, probability > 10 active at same time is less than .0004
 - No congestion in 99.96% of the time!





Packet Switching versus Circuit Switching

- Ten users in the system
 - Only one user active
- Download 10 Gbit video file
- Circuit switching
 - 10 seconds to download the file on a 1 Gbit/s circuit
- Packet switching
 - 1 second!
- Packet switching enables statistical multiplexing
 - Many users share the same resource
 - Not all of them can share it at the same time...
 - but we do not expect them to be all active at the same time





Packet switching versus circuit switching

- Is packet switching a *slam dunk winner*?
- Packet switching is well adapted for *bursty* data
 - Resource sharing
 - Simpler, no call setup
- Packet switching can cause excessive congestion
 - Delay and loss due to buffer overflow
- Protocols needed for reliable data transfer, congestion control!
- The restaurant reservation example!



Best Effort Internet Model: Packet Switching

- Internet is often described as *best effort* networks
- All packets are treated in the same fashion
 - The network undertakes its best effort to deliver every packet as quickly as it can
 - The network makes no undertaking to treat any class of packets preferentially to any other
- Best effort networking did not prevent the tremendous evolution of the Internet!



Conventional Approach to Communications: Circuit Switching

- The conventional approach to communications was developed with the public switched telephone network (PSTN)
 - Functionally rich network supporting very simple peripheral devices
- PSTN can support precisely clocked digital bit streams between two locations, with negligible delay variation and error levels
- PSTN performance is remarkable from an engineering perspective, but, from the perspective of the Internet architecture, somewhat irrelevant!



Best Effort Versus Conventional Telephony Approach

- The Internet adopted a revolutionary approach characterized as “dumb network, smart devices”
- The Internet assumes as little as possible about the characteristics (delivery success and order, throughput, delay, etc.) of the underlying transmission system
 - The Internet places all the functionality into the host computers at the edge of the network
- Expected service outcomes:
 - When there is no other network traffic, you can expect good quality of service
 - When the levels of traffic increase, you can expect congestion and loss



Best Effort Model, a Success?

- When you have a dropped call with PSTN, how do you react?
- When you have a dropped call with Internet applications for communication (Teams, WhatsApp, etc.), how do you react?
- How often do you encounter problems in receiving broadcast TV channels compared to the interruption of a YouTube or Netflix video?
- How often do you use legacy services (PSTN, TV broadcast, etc.) compared to Internet-based services?